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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	**ZeroTrustAI – AI-Powered Adaptive Identity and Access Management with Behavioral Risk Analytics**	
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ABSTRACT

ZeroTrustAI – AI-Powered Adaptive Identity and Access Management with Behavioural Risk Analytics is an intelligent cybersecurity platform that aims to improve access security by continuously verifying users, devices, and access requests based on the Zero Trust security principle. Unlike traditional authentication systems that mainly depend on passwords and fixed access rules, ZeroTrustAI analyses user behaviour, login patterns, device information, location, authentication history, and resource sensitivity to identify unusual activities and potential security threats. The system uses machine learning techniques such as XGBoost for risk prediction and Isolation Forest for anomaly detection, while SHAP provides understandable explanations for security decisions. Based on the calculated risk score, the system can dynamically allow access, request multi-factor authentication, restrict sensitive resources, or block suspicious sessions. Continuous authentication further helps monitor user behaviour even after login. An administrative dashboard provides security teams with a clear view of users, risk scores, suspicious activities, access decisions, and audit logs. Developed **using React.js, FastAPI, Python, PostgreSQL, Scikit-learn, XGBoost, SHAP, JWT, and Docker,** the proposed system provides a scalable and adaptive approach to identity and access management while demonstrating how AI and Adaptive Software Engineering can be applied to strengthen modern cybersecurity.

Signature of the Guide HOD